

Moazzam Maan

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EDUCATION

B.S. in Computer Science, B.S. in Physics, B.S. in Mathematics GPA: 4.00/4.00
Roanoke College, Salem, VA August 2023 — May 2027 (expected)

RESEARCH EXPERIENCE

Finding Largest Independent Sets in Circulant Graphs May 2024 — February 2026

Advisor: Dr. Anil Shende, Professor of Computer Science

- Used properties of independent sets in a circulant-graph model of a network of cellular towers to find an upper bound on the frequency separation between devices.
- Generalized these graphs into an infinite family and established bounds for the independence numbers of this broader family, while also determining the exact independence number in special cases.

High-Altitude Balloon January 2026 — May 2026

Advisor: Dr. Truong Le, Assistant Professor of Physics

- Worked in a nine-person interdisciplinary team to design, build, and launch a high-altitude balloon.
- Assembled and programmed the flight electronics, including sensors, an Arduino microcontroller, and onboard video equipment.
- Supported a successful launch in which the balloon reached approximately 27 km and landed safely, and worked on analyzing the collected data.

3D Wildfire Visualization February 2025 — September 2025

Advisor: Dr. Anil Shende, Professor of Computer Science

- Created a web-based software for visualizing 3D forest fire spread. Currently working on disseminating our product as open-source.
- Participated in the SERDP/ESTCP Virtual Wildfire Modeling and Immersive Training Prize Challenge and placed top 10 nationwide out of over 30 applicants.

Percolation Model of Spiral Galaxy Formation December 2025 — June 2026

Advisor: Dr. Truong Le, Assistant Professor of Physics

- Recreated the Seiden-Schulman gas-regulated galaxy formation model in Python, the first open-source reproduction.
- Added observational rotation curve data to recreate spiral galaxy morphology.
- Showed that 1–10 initial star-forming clusters can evolve into large-scale galaxy structure, matching early-galaxy observations such as the Firefly Sparkle.

Cops, Robber, and a Duck December 2025 — Present

Advisor: Dr. Michael Weselcouch, Assistant Professor of Mathematics

- Studying a variant of the pursuit-evasion graph theory game *Cops and Robbers*, where a temporary blocker changes which moves are available each turn.
- Developing graph constructions and strategies to compare this variant with the classical game as part of my Summer Scholars 2026 research project.

Interactive C++ Learning Environment

November 2025 — April 2026

Advisor: Dr. Durell Bouchard, Associate Professor of Computer Science

- Built a mini pedagogical C++ compiler and developed an interactive programming IDE in Minecraft.
- Designed programming lessons for CPSC-120 on sequential execution, loops, and conditional logic to support teaching in a game-based, interactive setting.

Statistical Analysis of Mario Kart 64 Race Data

January 2025 — Present

Advisor: Dr. Michael Weselcouch, Assistant Professor of Mathematics

- Created a Lua pipeline to extract real-time in-game MK64 data through BizHawk emulator memory.
- Conducted a semester-long research project collecting gameplay data from student races.
- Currently developing a SCORE Network-style educational module using Mario Kart 64 race data for statistics and data science classes.

PUBLICATIONS

1. **Structural Analysis of Largest Independent Sets in a Family of Circulant Graphs** (*with A. Shende*).
 - Proceedings of Joint Mathematics Meetings (JMM) (2025).
 - Proceedings of the Consortium for Computing Sciences in Colleges (CCSC) (2024).
 - Complete manuscript in preparation.
2. **Web-browser-based Visualization of Wildland Fire Data** (*with Y. Satynska and A. Shende*).
 - Proceedings of the Consortium for Computing Sciences in Colleges (CCSC) (2025).
3. **Percolation Models of Spiral Galaxy Formation** (*with T. Le, and C. Amberman*).
 - Proceedings of the 104th Annual Virginia Academy of Science Meeting, *Virginia Journal of Science* (2026, forthcoming).

TALKS AND PRESENTATIONS

1. **Interactive Web-Based Visualization of Wildland Fire Data**
 - Virtual Wildfire Modeling and Immersive Training Prize Challenge, Orlando, Florida, September 2025 (presentation).
 - Undergraduate Research Showcase at the Capitol, Virginia General Assembly, February 2026 (poster).
 - Mathematics, Computer Science and Physics Conversation Series, Roanoke College, November 2025 (talk).
 - CCSC Student Research Showcase, Mercer University, November 2025 (poster).
2. **Finding Largest Independent Sets in Families of Circulant Graphs**
 - Joint Mathematics Meetings (JMM) Student Research Showcase, Seattle Convention Center, January 2025 (poster).
 - National Conference on Undergraduate Research (NCUR), Greater Richmond Convention Center, April 2026 (poster).
 - Mathematics, Computer Science and Physics Conversation Series, Roanoke College, October 2024 (talk).
 - CCSC Student Research Showcase, Furman University, November 2024 (poster).
3. **Percolation Model of Spiral Galaxy Formation**
 - Virginia Academy of Science Annual Meeting (VAS), Christopher Newport University, 2026 (poster).

HONORS AND AWARDS

- **CCSC Research Competition:** First Place (2024), Third Place (2025).
- **VAS Annual Meeting:** Best Poster Presentation (2026).
- **CCSC Programming Competition:** Third Place (2025).
- **Roanoke College Research Showcase:** Third Place (2024).
- **SERDP Virtual Wildfire Modeling Challenge:** Top 10 national placement (2025).
- **Jane Ingram Computer Science Award** (2026).
- **Roanoke College Summer Scholar:** 2025, 2026.
- **Honor Societies:** Pi Mu Epsilon; Upsilon Pi Epsilon.

PROFESSIONAL EXPERIENCE

Virtual Reality Development Intern

February 2025 — July 2025

US Department of Agriculture - Forest Service

Missoula, MT (Remote)

- Developed an immersive and interactive web-based tool for visualizing output from QUIC-Fire, a state-of-the-art wildfire modeling software developed at Los Alamos National Laboratory.
- Attended bi-weekly meetings of the 3D/XR Community of Practice to exchange ideas with fire visualization developers and firefighters across the country.

Teaching Assistant for Math and Computer Science

August 2024 — December 2024

Computer Science Department, Math Department - Roanoke College

Salem, VA

- Assisted in teaching first-year courses in classes and labs.
- Provided feedback to the professor about suggested problems for the students in the class.

Subject Tutor for Computer Science, Math and Physics

Aug 2024 — Dec 2025

Center for Learning and Teaching - Roanoke College

Salem, VA

- Earned a Level 1 Certification through the College Reading and Learning Association.
- Provided tutoring sessions for 10+ courses.

Resident Advisor and Community Leader

August 2025 — Present

Residence Life and Housing - Roanoke College

Salem, VA

- Supervise and maintain the safety of 25+ residential students.
- Organize, manage, and execute bi-weekly events available to up to 200 residential students.

Pass On Hunger Treasurer

January 2024 — Present

Roanoke College Pass On Hunger - Food Distribution

Salem, VA

- Use the value of students' leftover meal swipes to donate ingredients to the Roanoke Rescue Mission.
- Communicate with 400 students to collect \$1,400 weekly donations and manage orders and drop-off.

SKILLS

- **Programming:** C++, Python, Java, JavaScript, Julia, R, Lua, PHP, SQL, Rust, Arduino/C.
- **Web and Software:** HTML/CSS, React, Three.js, A-Frame, GLSL, Maven, ANTLR, Git, Linux.
- **Technical Areas:** algorithms, data structures, graph theory, simulation, data analysis, data visualization, scientific computing, compiler/interpreter design, embedded systems.